Meteor Scatter Detector Prototype

Theory of operation:
The meteor detector prototype is proposed to consist of co-located radio transmitter and receiver The transmissions are directed overhead in pulses. The receiver would sense any return signal from meteor ionization trains between the pulses. A GPS receiver simultaneously captures time information. A computer captures active signals and stores them with time information for later processing.

Proposed Design

- 6m yagi
- Cross-elements for no polarity
- 50.07 MHz
- SWR <1.2:1
- PEP 200W
- Half-wave 10' elements
- 4-5 elements on boom

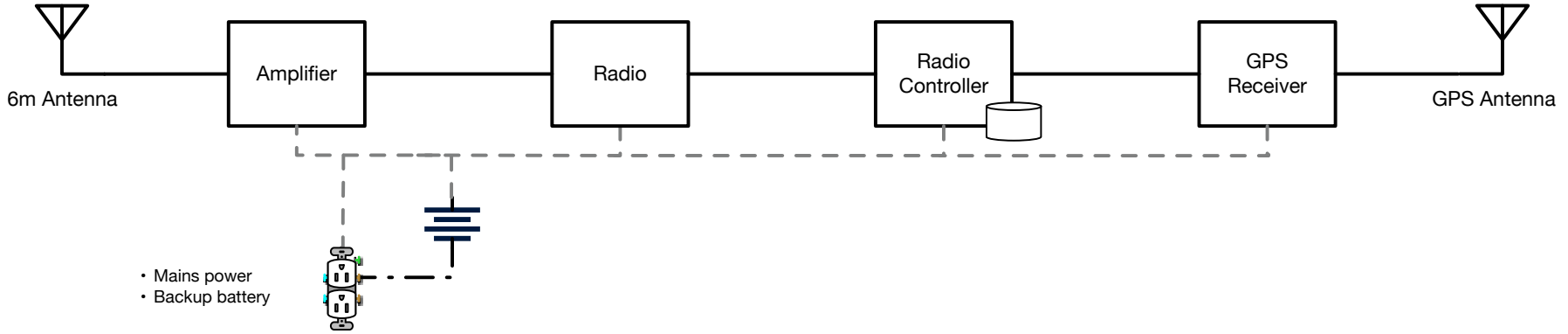
- 100W output
- 50% duty cycle

- 6m transceiver
- 5W output
- CAT controls
- I/Q stream output
- 1 sec send / 1 sec receive cycles

- Single Board Computer
- Radio cycle control
- I/Q Stream decimation
- GPS Decoding
- Timestamp extraction
- Data storage
- Wi-Fi hotspot
- Remote control

- Simple GPS receiver
- Module on radio controller

- Simple GPS antenna



Design Alternatives

<u>Better 6m antenna</u>
- More elements
- Full-wave 20' elements

<u>• One radio, no amp</u>
- 6m transceiver capable of 100W
- 50% duty cycle
- CAT controls
- I/Q stream output
- 1 sec send / 1 sec receive cycles

<u>Separate transmitter and receiver</u>
- 6m transmitter capable of 100W
- Separate receiver
- 50% duty cycle
- CAT controls
- I/Q stream output
- 1 sec send / 1 sec receive cycles

<u>Better computer</u>
- More powerful SBC

<u>Better GPS</u>
- Separate GPS receiver
- Multi-band capable
- On-board GPS decoding
- On-board timestamp extraction

<u>Better GPS antenna</u>
- Multi-band GPS antenna

- NOTES:
- Not designed for extended outdoor use or inclement weather
- Duration dependent on power supply / battery
- Storage limitations dictated by size of memory card on radio controller

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